

AUTHOR DECLARATIONS & SUBMISSION STATEMENTS

Funding Statement

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. The Project Setu initiative was conducted independently at AeroMyne.

Declaration of Competing Interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Author Contributions (CRediT format)

Prashant Suresh Kamble: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Project administration.

Data and Code Availability Statement

The complete multi-physics simulation framework, ANSYS MAPDL macro routines (.apdl), Ansys Lumerical FDTD model scripts (.lsf), and numerical validation datasets generated during this study are archived and available in the Project Setu numerical repository.